**Part 1:**
**NSO workflow from device onboarding → config → rollback → templates**

---

```
Login
   ↓
Discover Devices
   ↓
Connect Devices
   ↓
Synchronize Configuration (sync-from: Device → NSO CDB)
   ↓
Create and Manage Device Groups
   ↓
Modify Configuration (Candidate Configuration)
   ↓
Validate Changes (Dry-run / Preview)
   ↓
Commit Configuration (Apply to Devices)
   ↓
Rollback Changes (if required)
   ↓
Create and Use Templates (Automation)
```

---

### ✅ ✅ Phase 1: Login & Device Discovery

🔹 **Command**

ncs_cli -C -u admin

✅ **Purpose**

- Log into NSO CLI
- -C → CLI mode
- -u admin → admin user

---

🔹 **Command**

show devices list

✅ **Purpose**

- Lists all onboarded devices
- Shows:
    - Device name
    - NED mapping (vendor type)

👉 Confirms devices are correctly added in NSO

---

### ✅ ✅ Phase 2: Device Connectivity & Sync

🔹 **Command**

devices connect

✅ **Purpose**

- Establish connection (SSH/NETCONF) to devices

  ✔️ Required before configuration

---

🔹 **Command**

show running-config devices device edge-sw0 config

✅ **Before sync**

- Shows minimal or no config

  👉 Because NSO has not yet pulled device config

---

🔹 **Command**

devices sync-from

✅ **Purpose**

- Pulls config from device → NSO CDB

👉 Synchronizes:

Device → NSO

---

🔹 **Command**

show running-config devices device edge-sw0 config

✅ **After sync**

- Shows full config like:
  - Interfaces
  - BGP
  - Policies

👉 Now NSO knows actual device state

---

✅ ✅ **Phase 3: Device Grouping**

🔹 **Commands**

config

devices device-group XR-DEVICES

devices device-group NXOS-DEVICES

devices device-group IOS-DEVICES

devices device-group ASA-DEVICES

devices device-group ALL

✅ **Purpose**

- Create logical device groups

  👉 Useful for:

- Bulk operations
- Service deployment

---

🔹 **Command**

commit and-quit
` `

✅ **Purpose**

- Save configuration
- Exit config mode

---

🔹 **Command**

devices device-group IOS-DEVICES check-sync

✅ **Purpose**

- Verify NSO vs device sync

Output:

- ✅ in-sync → OK
- ❌ out-of-sync → mismatch

---

✅ ✅ **Phase 4: Configuration Inspection**

🔹 **Commands**

show running-config devices device internet-rtr0

show running-config devices device internet-rtr0 config | display json

✅ **Purpose**

- View config in:
  - CLI format ✅
  - JSON format ✅

---

🔹 **Command**

show running-config devices device dist-sw0 interface port-channel

✅ **Purpose**

- Filter specific configuration

  👉 Helps in troubleshooting

---

✅ ✅ **Phase 5: Device Configuration**

🔹 **Enter config mode**

config

devices device dist-sw0 config
` `

---

🔹 **VLAN Creation**

vlan 42

name TheAnswer

✅ Creates VLAN

---

🔹 **Interface Configuration**

interface Vlan42

ip address 10.42.42.42/24

\`\`

✅ Configures SVI interface

---

✅ ✅ **Phase 6: Validation (Before Commit)**

🔹 **Command**

show configuration

✅ Shows:

- Pending changes in NSO (candidate)

---

🔹 **Command**

show configuration | display xml

\`\`

✅ Shows:

- YANG/XML structure

---

🔹 **Command**

commit dry-run outformat native

✅ **Purpose**

- Preview actual device CLI

👉 Example output:

vlan 42

interface Vlan42

ip address 10.42.42.42

✅ No changes applied yet

---

✅ ✅ **Phase 7: Apply Configuration**

🔹 **Command**

commit

✅ **Purpose**

- Apply config to device

👉 Flow:

Candidate → FASTMAP → NED → Device

---

✅ ✅ **Phase 8: Rollback**

🔹 **Command**

show configuration rollback changes

✅ Shows diff:

no vlan 42

no interface Vlan42

---

🔹 **Command**

rollback configuration

✅ Reverts candidate changes

---

🔹 **Command**

commit

✅ Applies rollback to device

👉 Removes VLAN

---

✅ ✅ **Phase 9: Template Creation**

🔹 **Command**

devices template SET-DNS-SERVER

✅ Creates reusable template

---

🔹 **Command**

ned-id cisco-nx-cli-3.0

✅ Binds template to NXOS devices

---

🔹 **Command**

ip name-server 208.67.x.x

✅ Add DNS server config

---

🔹 **Command**

show configuration

commit

✅ Saves template

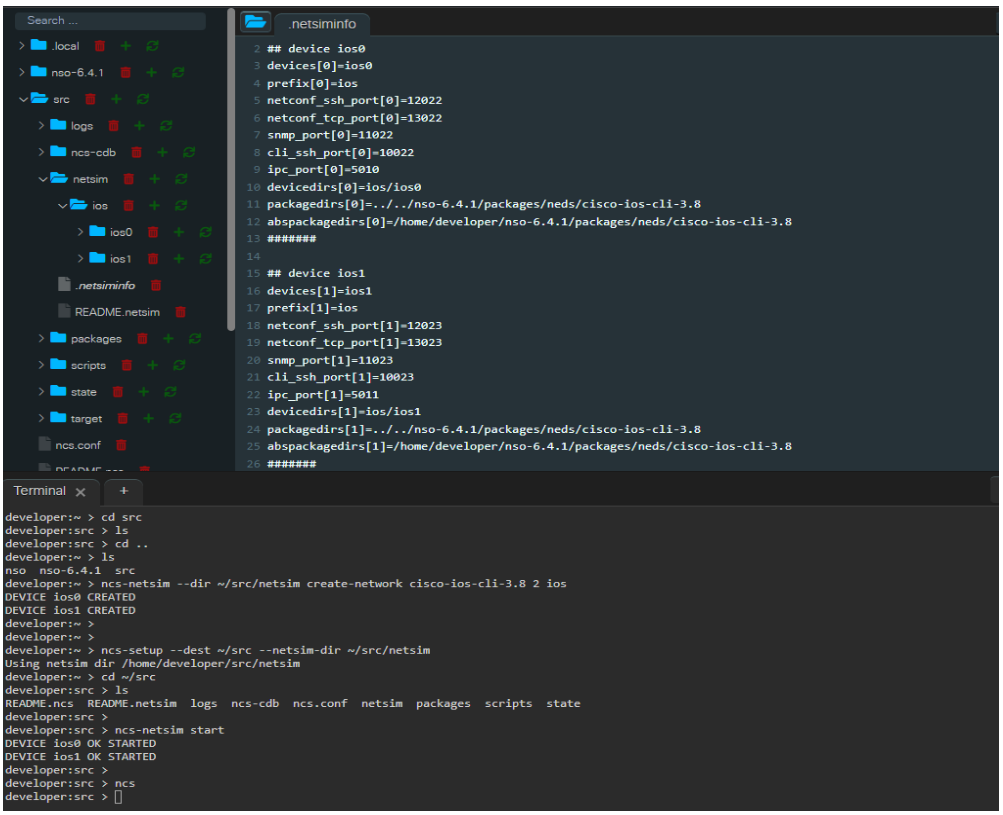
---

PART 2:

**This workflow demonstrates the complete Cisco NSO lifecycle: environment setup, device synchronization, configuration management, service creation, modification, deletion, and automation using service packages.**

Setup → Start NSO → Login → Sync → Verify
→ Config Mode → View → Modify → Dry-run → Commit
→ Create Package → Load → Verify
→ Create Service → Preview → Commit
→ Modify Service → Commit → Re-deploy
→ Delete Service → Commit → Verify Cleanup

---



```
Terminal  ×   +
developer:~ > cd src
developer:src > ls
developer:src > cd ..
developer:~ > ls
nso  nso-6.4.1  src
developer:~ > ncs-netsim --dir ~/src/netsim create-network cisco-ios-cli-3.8 2 ios
DEVICE ios0 CREATED
DEVICE ios1 CREATED
developer:~ >
developer:~ >
developer:~ > ncs-setup --dest ~/src --netsim-dir ~/src/netsim
Using netsim dir /home/developer/src/netsim
developer:~ > cd ~/src
developer:src > ls
README.ncs  README.netsim  logs  ncs-cdb  ncs.conf  netsim  packages  scripts  state
developer:src >
developer:src > ncs-netsim start
DEVICE ios0 OK STARTED
DEVICE ios1 OK STARTED
developer:src >
developer:src > ncs
developer:src >
developer:src > ncs_cli -u admin -C

admin connected from 127.0.0.1 using console on devpod-8929260738932363949-6c656c4878-kcvvm
admin@ncs#
admin@ncs# devices sync-from
sync-result {
    device ios0
    result true
}
sync-result {
    device ios1
    result true
}
admin@ncs#
admin@ncs# devices device ios1 check-sync
result in-sync
admin@ncs#
```

```
  interface Loopback0
   no shutdown
  exit
  interface FastEthernet0/0
   no shutdown
  exit
  interface FastEthernet1/0
   no shutdown
  exit
  interface FastEthernet1/1
   no shutdown
  exit
  no spanning-tree optimize bpdu transmission
  router bgp 64512
   aggregate-address 10.10.10.1 255.255.255.251
  !
 !
!
admin@ncs(config)#
```

```
admin@ncs(config)#
admin@ncs(config)# show full-configuration devices device ios1 config | include bgp
   bgp next-hop Loopback1
  router bgp 64512
admin@ncs(config)#
```

---

Here is your **same grid (unchanged order) + clear description below each grid** ✅

👉 Clean and ready for documentation/notes

```
Terminal  ×   +
developer:~ > cd src
developer:src > ls
developer:src > cd ..
developer:~ > ls
nso  nso-6.4.1  src
```

## ✅ ✅ Netsim Setup

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 1 | ncs-netsim create-network | Create simulated devices |
| 2 | ncs-setup | Setup NSO environment |
| 3 | cd ~/src | Enter workspace |
| 4 | ls | Verify files |
| 5 | ncs-netsim start | Start devices |

### ✅ Description:

This phase sets up a virtual lab environment using **NetSim**. It creates simulated IOS devices, prepares the NSO workspace, and starts the virtual network so NSO can interact with devices without real hardware.

```
developer:~ > ncs-netsim --dir ~/src/netsim create-network cisco-ios-cli-3.8 2 ios
DEVICE ios0 CREATED
DEVICE ios1 CREATED
developer:~ >
developer:~ >
developer:~ > ncs-setup --dest ~/src --netsim-dir ~/src/netsim
Using netsim dir /home/developer/src/netsim
developer:~ > cd ~/src
developer:src > ls
README.ncs  README.netsim  logs  ncs-cdb  ncs.conf  netsim  packages  scripts  state
developer:src >
developer:src > ncs-netsim start
DEVICE ios0 OK STARTED
DEVICE ios1 OK STARTED
developer:src >
```

---

## ✅ ✅ Start NSO & Login

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 6 | ncs | Start the Cisco NSO server |
| 7 | ncs_cli -u admin -C | Connect to NSO CLI |

### ✅ Description:

This step starts the NSO engine and allows the user to log in using the CLI. NSO loads the CDB, packages, and begins orchestration services.

```
developer:src > ncs
developer:src >
developer:src > ncs_cli -u admin -C

admin connected from 127.0.0.1 using console on devpod-8929260738932363949-6c656c4878-kcvvm
admin@ncs#
```

---

## ✅ ✅ Device Sync & Verification

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 8 | devices sync-from | Sync device configuration into NSO CDB |
| 9 | devices device ios1 check-sync | Verify NSO and device are in-sync |

✅ **Description:**

NSO pulls the actual device configuration into its internal database (CDB) and verifies that both NSO and device configurations match.

```
admin@ncs# devices sync-from
sync-result {
    device ios0
    result true
}
sync-result {
    device ios1
    result true
}
admin@ncs#
admin@ncs# devices device ios1 check-sync
result in-sync
admin@ncs#
```

---

## ✅ ✅ Enter Configuration Mode

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 10 | config | Enter configuration mode to make changes |

✅ **Description:**

Switches NSO into **candidate configuration mode**, where all changes are staged temporarily before being committed to devices.

```
admin@ncs#
admin@ncs# config
Entering configuration mode terminal
admin@ncs(config)#
```

---

## ✅ ✅ View Device Configuration

### 🔹 Grid

| Step | Command | Purpose | |
|---|---|---|---|
| 11 | show full-configuration devices device ios1 config | nomore | Display full config |
| 12 | show full-configuration devices device ios1 config | include bgp | Filter BGP |
| 13 | devices device ios0 config | Enter device config mode | |
| 14 | show full-configuration | nomore | Show full current device config |

✅ **Description:**

Used to inspect device configurations stored in NSO. It supports full view, filtered view, and device-specific navigation.

```
admin@ncs(config)#
admin@ncs(config)# show full-configuration devices device ios1 config | nomore
devices device ios1
 config
  tailfned device netsim
  tailfned police cirmode
  no service password-encryption
  no cable admission-control preempt priority-voice
  no cable qos permission create
  no cable qos permission update
  no cable qos permission modems
  ip source-route
  no ip cef
  ip vrf my-forward
   bgp next-hop Loopback1
  !
  no ip forward-protocol nd
  no ip http server
  no ip http secure-server
  ip community-list 1 permit
  ip community-list 2 deny
  ip community-list standard s permit
  no ipv6 source-route
  no ipv6 cef
  class-map match-all a
  !
  class-map match-all cmap1
   match mpls experimental topmost 1
   match packet length max 255
   match packet length min 2
   match qos-group 1
  !
  policy-map a
  !
  policy-map map1
   class c1
    drop
    estimate bandwidth delay-one-in 500 milliseconds 100
    priority percent 33
   !
  !
  interface Loopback0
   no shutdown
  exit
  interface FastEthernet0/0
   no shutdown
  exit
  interface FastEthernet1/0
   no shutdown
  exit
  interface FastEthernet1/1
   no shutdown
  exit
  no spanning-tree optimize bpdu transmission
  router bgp 64512
   aggregate-address 10.10.10.1 255.255.255.251
  !
 !
!
admin@ncs(config)#
```

```
admin@ncs(config)# show full-configuration devices device ios1 config | include bgp
   bgp next-hop Loopback1
  router bgp 64512
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)# devices device ios0 config
admin@ncs(config-config)#
admin@ncs(config-config)#
admin@ncs(config-config)# show full-configuration | nomore
devices device ios0
 config
  tailfned device netsim
  tailfned police cirmode
  no service password-encryption
  no cable admission-control preempt priority-voice
  no cable qos permission create
  no cable qos permission update
  no cable qos permission modems
  ip source-route
  no ip cef
  ip vrf my-forward
   bgp next-hop Loopback1
  !
  no ip forward-protocol nd
  no ip http server
  no ip http secure-server
  ip community-list 1 permit
  ip community-list 2 deny
  ip community-list standard s permit
  no ipv6 source-route
```

```
  no ipv6 cef
  class-map match-all a
  !
  class-map match-all cmap1
   match mpls experimental topmost 1
   match packet length max 255
   match packet length min 2
   match qos-group 1
  !
  policy-map a
  !
  policy-map map1
   class c1
    drop
    estimate bandwidth delay-one-in 500 milliseconds 100
    priority percent 33
   !
  !
  interface Loopback0
   no shutdown
  exit
  interface FastEthernet0/0
   no shutdown
  exit
  interface FastEthernet1/0
   no shutdown
  exit
  interface FastEthernet1/1
   no shutdown
  exit
  no spanning-tree optimize bpdu transmission
  router bgp 64512
   aggregate-address 10.10.10.1 255.255.255.251
  !
 !
!
admin@ncs(config-config)#
admin@ncs(config-config)#
```

## ✅ ✅ Device Configuration Changes

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 15 | ios:hostname nso.cisco.com | Set hostname |
| 16 | top | Move to root |

| Step | Command | Purpose |
|---|---|---|
| 17 | show configuration | Show pending changes |
| 18 | devices device * config ios:enable password magic | Set enable password |

✅ **Description:**

This stage modifies the **candidate configuration**. Changes like hostname and password are staged but not yet applied to devices.

```
admin@ncs(config-config)# ios:hostname nso.cisco.com
admin@ncs(config-config)#
admin@ncs(config-config)#
```

```
admin@ncs(config-config)#
admin@ncs(config-config)# top
admin@ncs(config)#
```

```
admin@ncs(config)# show configuration
devices device ios0
 config
  hostname nso.cisco.com
 !
!
admin@ncs(config)#
```

```
admin@ncs(config)#
admin@ncs(config)# devices device * config ios:enable password magic
admin@ncs(config-config)#
```

---

## ✅ ✅ Validation & Apply

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 19 | commit dry-run outformat native | Preview CLI |
| 20 | commit | Apply configuration |

✅ **Description:**

Dry-run shows the exact device commands before execution. Commit pushes changes to devices using NEDs (southbound).

```
admin@ncs(config-config)#
admin@ncs(config-config)# show configuration
devices device ios0
 config
  hostname nso.cisco.com
  enable password magic
 !
!
devices device ios1
 config
  enable password magic
 !
!
admin@ncs(config-config)#
admin@ncs(config-config)#
admin@ncs(config-config)# commit dry-run outformat native
native {
    device {
        name ios0
        data hostname nso.cisco.com
             enable password magic
    }
    device {
        name ios1
        data enable password magic
    }
}
admin@ncs(config-config)#
admin@ncs(config-config)#
admin@ncs(config-config)#
admin@ncs(config-config)# commit
Commit complete.
admin@ncs(config-config)# 
```

```
admin@ncs(config-config)#
admin@ncs(config-config)# commit
Commit complete.
admin@ncs(config-config)# end
admin@ncs#
admin@ncs#
admin@ncs# exit
developer:src > 
```

---

## ✅ ✅ Create Service Package

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 21 | ncs-make-package ... | Create service package |
| 22 | make | Build package |
| 23 | packages reload | Load package |

### ✅ Description:

Creates a **service package** with YANG, templates, and structure. This enables automation and reusable service definitions.

```
developer:src > ncs-make-package --service-skeleton template --dest ~/src/packages/simple-service simple-service
developer:src > printf 'module simple-service {
>   namespace "http://com/example/simpleservice";
>   prefix simple-service;
>   import tailf-ncs { prefix ncs; }
>   list simple-service {
>     key name;
>     uses ncs:service-data;
>     ncs:servicepoint simple-service;
>     leaf name {
>       type string;
>     }
>     leaf device {
>       type leafref {
>         path "/ncs:devices/ncs:device/ncs:name";
>       }
>     }
>     leaf secret {
>       type string;
>     }
>   }
> }
> ' > ~/src/packages/simple-service/src/yang/simple-service.yang
developer:src > printf '<config-template xmlns="http://tail-f.com/ns/config/1.0" servicepoint="simple-service">
>   <devices xmlns="http://tail-f.com/ns/ncs">
>     <device>
>       <name>{./device}</name>
>       <config>
>         <enable xmlns="urn:ios">
>           <password>
>             <secret>{./secret}</secret>
>           </password>
>         </enable>
>       </config>
>     </device>
>   </devices>
> </config-template>
> ' > ~/src/packages/simple-service/templates/simple-service-template.xml
developer:src > make -C ~/src/packages/simple-service/src
make: Entering directory '/home/developer/src/packages/simple-service/src'
mkdir -p ../load-dir
/home/developer/nso-6.4.1/bin/ncsc `ls simple-service-ann.yang  > /dev/null 2>&1 && echo "-a simple-service-ann.yang"` \
        --fail-on-warnings \
         \
        -c -o ../load-dir/simple-service.fxs yang/simple-service.yang
make: Leaving directory '/home/developer/src/packages/simple-service/src'
developer:src > echo "packages reload" | ncs_cli -C -u admin

>>> System upgrade is starting.
>>> Sessions in configure mode must exit to operational mode.
>>> No configuration changes can be performed until upgrade has completed.
>>> System upgrade has completed successfully.
reload-result {
    package cisco-ios-cli-3.8
```

```
    result true
}
reload-result {
    package simple-service
    result true
}
developer:src >
```

---

## ✅ ✅ Verify Packages

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 24 | show packages package oper-status | Verify packages |

### ✅ Description:

Checks whether all service and NED packages are successfully loaded and operational in NSO.

```
developer:src >
developer:src > ncs_cli -C -u admin

User admin last logged in 2026-05-18T10:23:01.892613+00:00, to devpod-8929260738932363949-6c656c4878-kcvvm, from 127.0.0.1 using cli-console
admin connected from 127.0.0.1 using console on devpod-8929260738932363949-6c656c4878-kcvvm
admin@ncs#
admin@ncs#
admin@ncs# show packages package oper-status
packages package cisco-ios-cli-3.8
 oper-status up
packages package simple-service
 oper-status up
admin@ncs#
```

---

## ✅ ✅ Create Service Instance

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 25 | simple-service test1 ... | Create service instance |
| 26 | show configuration | Verify input |
| 27 | commit dry-run | FASTMAP preview |
| 28 | commit dry-run outformat native | CLI preview |
| 29 | commit | Apply service |

✅ **Description:**

Defines a service instance using input parameters. NSO uses FASTMAP to convert intent into device configuration and applies it.

```
admin@ncs# config
Entering configuration mode terminal
admin@ncs(config)#
admin@ncs(config)# simple-service test1 device ios0 secret mypasswd
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# show configuration
simple-service test1
 device ios0
 secret mypasswd
!
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# commit dry-run
cli {
    local-node {
        data +simple-service test1 {
             +    device ios0;
             +    secret mypasswd;
             +}
              devices {
                  device ios0 {
                      config {
                          enable {
                              password {
             -                    secret magic;
             +                    secret mypasswd;
                              }
                          }
                      }
                  }
              }
    }
}
admin@ncs(config-simple-service-test1)#
```

```
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# commit dry-run outformat native
native {
    device {
        name ios0
        data enable password mypasswd
    }
}
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# commit
Commit complete.
admin@ncs(config-simple-service-test1)#
```

---

## ✅ ✅ Modify Service

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 30 | secret securepasswd | Modify parameter |
| 31 | commit dry-run | Preview changes |
| 32 | commit | Apply changes |
| 33 | simple-service test1 get-modifications | Show diff |
| 34 | simple-service test1 re-deploy | Reapply service |

✅ **Description:**

Updates an existing service. NSO calculates delta changes and applies only the differences using FASTMAP.

```
admin@ncs(config-simple-service-test1)# secret securepasswd
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# commit dry-run
cli {
    local-node {
        data  simple-service test1 {
             -    secret mypasswd;
             +    secret securepasswd;
              }
              devices {
                  device ios0 {
                      config {
                          enable {
                              password {
             -                    secret mypasswd;
             +                    secret securepasswd;
                              }
                          }
                      }
                  }
              }
    }
}
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# commit
Commit complete.
admin@ncs(config-simple-service-test1)#
admin@ncs(config-simple-service-test1)# top
admin@ncs(config)# simple-service test1 get-modifications
cli {
    local-node {
        data  devices {
                   device ios0 {
                       config {
                           enable {
                               password {
              -                    secret magic;
              +                    secret securepasswd;
                               }
                           }
                       }
                   }
               }

    }
}
admin@ncs(config)# simple-service test1 re-deploy
admin@ncs(config)#
System message at 2026-05-18 10:33:36...
Commit performed by admin via console using cli.
admin@ncs(config)#
```

---

## ✅ ✅ Delete Service

### 🔹 Grid

| Step | Command | Purpose |
|---|---|---|
| 35 | no simple-service test1 | Delete service |
| 36 | commit dry-run | Preview removal |
| 37 | commit | Remove config |

### ✅ Description:

Deletes the service and automatically removes related configuration from devices.

```
admin@ncs(config)# no simple-service test1
admin@ncs(config)#
admin@ncs(config)# commit dry-run
cli {
    local-node {
        data -simple-service test1 {
             -    device ios0;
             -    secret securepasswd;
             -}
              devices {
                  device ios0 {
                      config {
                          enable {
                              password {
             -                    secret securepasswd;
             +                    secret magic;
                              }
                          }
                      }
                  }
              }
    }
}
admin@ncs(config)#
admin@ncs(config)# commit
Commit complete.
admin@ncs(config)#
```

---

### ✅ ✅ Post Deletion Behavior

🔹 **Grid**

| Step | Command | Purpose |
|---|---|---|
| 38 | simple-service test1 get-modifications | Fails |
| 39 | simple-service test1 re-deploy | Fails |
| 40 | commit dry-run | No changes |
| 41 | commit | No operation |

### ✅ Description:

Confirms that the service is completely removed. No further operations are possible since the service instance no longer exists.

```
admin@ncs(config)# top
admin@ncs(config)# top
admin@ncs(config)# simple-service test1 get-modifications
Error: List instance does not exist
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)# simple-service test1 re-deploy
Error: List instance does not exist
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)# no simple-service test1
admin@ncs(config)#
admin@ncs(config)#
admin@ncs(config)# commit dry-run
% No modifications to commit.
admin@ncs(config)# commit
% No modifications to commit.
admin@ncs(config)#
```

---

**Summary:**

Setup (NetSim Environment)

↓

```
Start NSO Server
  ↓
Login to NSO CLI
  ↓
Sync Devices (Device → NSO CDB)
  ↓
Verify Device Sync Status
  ↓
Enter Configuration Mode (Candidate)
  ↓
View Device Configuration
  ↓
Modify Candidate Configuration
  ↓
Preview Changes (Dry-run)
  ↓
Commit Changes (Apply to Devices)
  ↓
Create Service Package (YANG + Template)
  ↓
Build & Load Package (Package Manager)
  ↓
Verify Package Status
  ↓
Create Service Instance (Input Intent)
  ↓
Validate & Preview (FASTMAP Diff)
  ↓
Apply Service (Commit)
  ↓
Modify Service (Update Parameters)
  ↓
Preview Changes (FASTMAP Diff)
  ↓
Apply Updated Service
  ↓
Re-deploy Service (if needed)
  ↓
Delete Service Instance
  ↓
Preview Removal Changes
```

↓

Commit Deletion (Restore Device State)

↓

Verify Service Removal

↓

Confirm No Pending Changes

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